Carbon Markets

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Global Carbon Markets: The State of Play (2025–26)

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Carbon Pricing Instruments

Across ~55 national & 44 sub-national jurisdictions covering ~28% of global GHG emissions (14.7 GtCO2e)

$950bn

Compliance Market Turnover (2024)

EU ETS, China ETS, UK ETS, Korea K-ETS dominating volume

$1.7bn

Voluntary Market Value

Recovering post-integrity reset; ICVCM Core Carbon Principles emerging as quality benchmark

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Mega Forces Reshaping Markets

- CBAM enforcement
- Article 6 operationalization
- New ETS launches (India, Brazil, Turkey, Vietnam)

1.51 $BtCO_2e$

Estimation of offset requirements in CORSIA

Price scenarios for CORSIA-eligible carbon credits:

- Phase I (2024–2026): $18–$51 per tonne
- Phase II (2027–2035): $27–$91 per tonne

We are entering the era of regulated, priced, and verified carbon.

Comparative Global ETS Benchmarking

Parameter	EU ETS	China ETS	India CCTS
Launch Year	2005 (Phase IV since 2021)	2021	2023 (rules notified 2025)
Mechanism	Cap-and-Trade (absolute cap)	Baseline-and-Credit (intensity)	Baseline-and-Credit (intensity)
Coverage	~40% of EU GHG Maritime, Domestic Aviation, Industry, Power	~40% of China GHG; Industry and Power	490 industrial units; expansion planned
Price (~2025)	€65–80/tCO_2e	~$11–14/t$CO_2$e	To be discovered on exchanges
Allocation	Auctioning dominant (57%+)	100% free (Output-based Benchmarking)	Intensity-based baseline-and-credit; legally binding GEI targets (2025–27); over-achievers generate CCCs
Offset Usage	The use of offsets is not allowed (Banned since Phase IV)	CCER (relaunched 2024)	Currently, the use of offsets is not allowed
CBAM Linkage	CBAM Phase 2 from Jan 2026	Potential retaliation	CCTS as implicit CBAM defence

Next steps for India

- Strengthen MRV comparability with international standards
- Ensure registry + exchange plumbing is ready for liquidity
- Develop guardrails for credit integrity
- Develop CBAM strategy for exporters
- Calibrate ambition over time

Carbon Market Taxonomy

Compliance Carbon Markets (CCMs)

- **Cap-and-Trade / ETS**: Government sets absolute cap → entities trade (EU ETS model)
- **Baseline-and-Credit**: Intensity benchmarks → over-performers generate credits (India CCTS model)
- **Carbon Tax**: Fixed price/tCO_2 — not market-based (Singapore, Canada)

Voluntary Carbon Markets (VCMs)

- **Carbon Market Standards**: Verra (VCS), Gold Standard, Global Carbon Council (GCC)
- **Integrity standards:** ICVCM Core Carbon Principles, VCMI Claims Code of Practice
- **Demand**: Corporate net-zero pledges, SBTi targets, CORSIA for aviation, ESG investor pressure

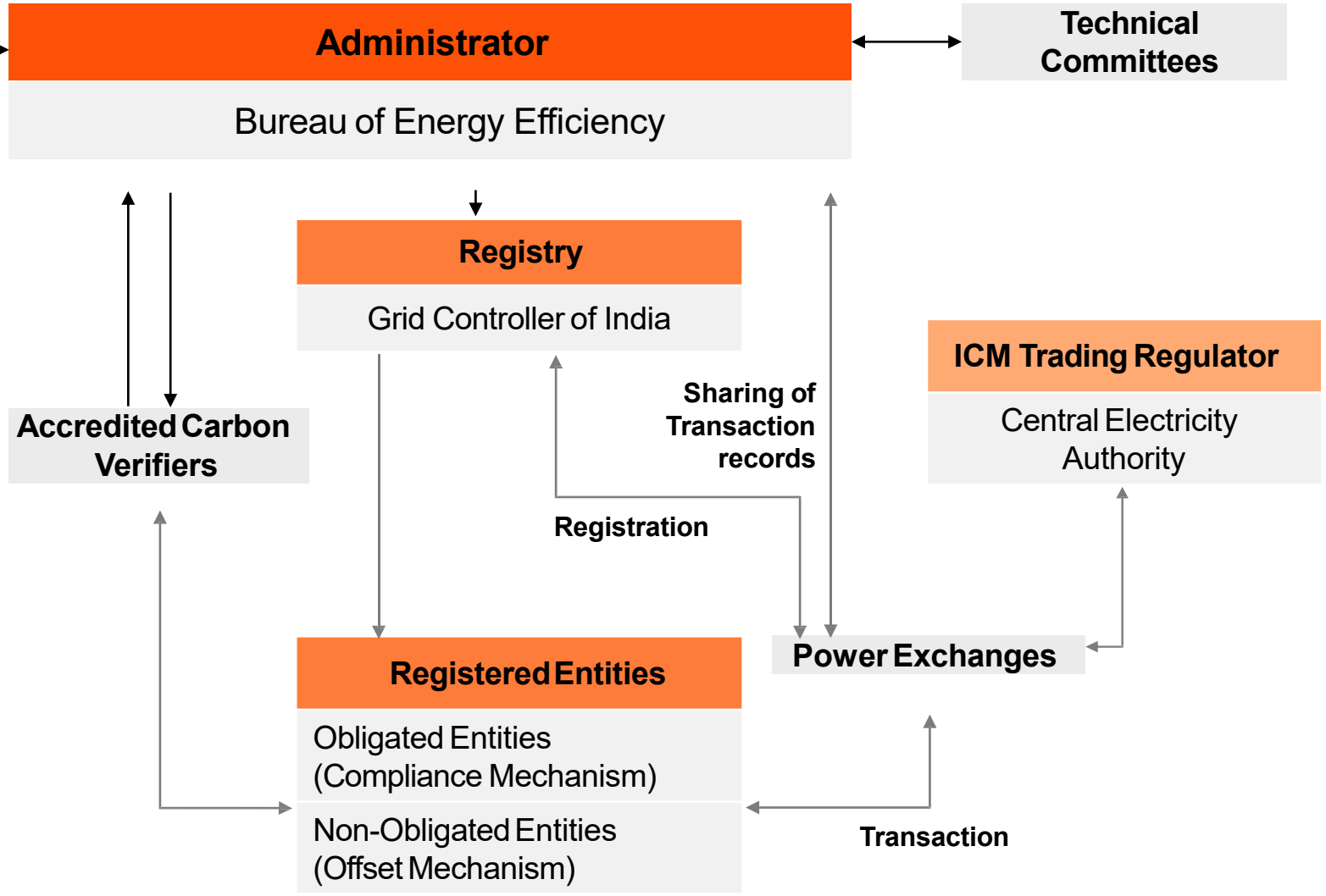
The Emerging Convergence (2025–30)

- India's offset mechanism bridges compliance & voluntary
- Article 6.2 and Article 6.4 ITMOs create sovereign layer above both
- CBAM creates cross-border compliance pull
- ISSB/BRSR disclosure turns voluntary into de facto regulatory

The wall between compliance and voluntary is dissolving. Design your carbon strategy for convergence, not silos.

India's CCTS Institutional Architecture

National Steering Committee for Indian Carbon Market (NSC-ICM)	
Co-Chaired by – Secretary, MoEFCC and Secretary MoP	
Members	
Ministry of Finance	Ministry of New and Renewable Energy
Ministry of Steel	Ministry of Power
Ministry of Civil Aviation	Ministry of Ports, Shipping and Waterways
Ministry of Environment forest and Climate Change	Ministry of Chemical and Fertilizers
Ministry of Agriculture and Farmers Welfare	Central Electricity Authority
Grid Controller of India Ltd.	Member Secretary, Director General - BEE
NITI Aayog	Department of Environment or State (2) Expert Members - (2)



Compliance Market Deep-Dive: How It Actually Works

The CCTS Compliance Cycle — Step by Step

1. Target Setting

BEE assigns GHG emission intensity benchmarks (tCO_2e/unit output) to 282 obligated entities

2. Project Registration

Obligated entities open accounts on the ICM Registry

3. Monitoring plan and GRAP

Submit a Monitoring Plan and GHG Reduction Action Plan to BEE

4. Monitoring and Recording

Plant-level data captured through the FY as per approved MRV procedures

5. Third-Party Verification

An Accredited Carbon Verification Agency (ACVA) verifies performance and issues a verification report

6. Credit Generation / Deficit

Over-performers → generate CCCs (surplus). Under-performers → must acquire CCCs.

7. Trading

CCCs traded on designated exchanges (IEX, PXIL, HPX)

8. Compliance or Penalty

Surrender CCCs to meet target OR pay financial penalty for non-adherence

The Offset Ecosystem: Approved Activities & Demand Creation

Building the Voluntary-Compliance Bridge

India's CCTS Offset mechanism

Detailed procedure

Offset Mechanism is formalized by BEE's Detailed Procedure (Mar 2025)

Sectoral scope build-out

- Phase 1: energy, industry, waste, agriculture, forestry, transport
- Phase 2: construction, fugitive emissions, solvent use, CCUS

Early methodology slate

Centre approved 9 methodologies under energy, industry, Waste, Agriculture and Forestry sectors

BEE approval = regulatory endorsement → higher integrity perception than pure voluntary market credits. Credits are Article 6.2 export-ready and CBAM-defense-relevant

Global integrity anchors (for premium demand)

ICVCM: Core Carbon Principles (CCPs) + growing list of CCP-approved methodologies are becoming the quality bar buyers seek

VCMI: Claims Code v3.0 (Apr 2025) guides credible corporate use/claims and alignment with science-based targets.

Aviation Demand: CORSIA only accepts units from ICAO-approved programs

Article 6: the pathway to cross-border trade

COP decisions: Authorizations, first transfer sequencing, reporting/registry interoperability—key to export ITMOs

Bridge principle: Exports for international use (A6.2/6.4/ CORSIA) require host authorization and corresponding adjustments to avoid double counting.

Demand Creation

- Corporate net-zero & disclosure drive domestic VCM demand; CCP/VCMI-aligned credits command price premia.
- CORSIA route for eligible Indian projects via approved programs; airlines' compliance demand persists through 2024–26 (and beyond)
- Public procurement pilots & sector coalitions (cement/steel/value-chains) can aggregate early demand while compliance rules mature.

EU CBAM Implications for Indian Industry

The Carbon Border Tax That Changes Trade Economics — and the FTA Calculus

What changes from 1 Jan 2026?

- **CBAM's definitive phase is live**: EU Importers must purchase CBAM certificates linked to the EU ETS carbon price
- Only authorized CBAM Declarants can import
- Sale/surrender of certificates for 2026 imports begins in 2027 under EU rules.

Who is exposed?

- **Sectors:** Iron & steel, aluminium, cement, fertilizers, electricity, hydrogen
- Analyses indicate Indian steel & aluminium face the sharpest shock
- India's exports to EU in 2025:
  - Iron & Steel - €4.2 Bn
  - Aluminium - €800 Mn

Way ahead for Indian firms

- Get installation-level MRV ready at process/installation level
- Nominate/align with EU importers as Authorized CBAM Declarants
- Model landed-cost scenarios at EUA price ranges and plan decarbonisation levers

CCTS price signal & MRV can underpin embedded-emissions disclosure and enable deduction of any domestic carbon price under CBAM—provided payments and accounting are robustly evidenced.

The FTA Negotiation Leverage

- India's position: If CCTS is a credible domestic carbon price, EU should deduct equivalent from CBAM liability
- Article 25 of CBAM Regulation allows recognition — but requires 'effective' carbon price covering same sectors/gases
- CCTS operationalization is not just climate policy — it is trade policy

Article 6 & Cross-Border Carbon Trade

Countries can voluntarily cooperate to achieve their NDCs in whatever way they wish

Article 6.2

- Countries can trade Article 6 units through bilateral or multilateral agreements - or even through unilateral authorizations
- Enables a seller country to trade units to attract investments and access to technologies that might not be available domestically
- ITMOs with corresponding adjustments
- India – Japan Joint Crediting Mechanism MOU signed
- **Potential cooperative approaches:** Singapore, Sweden, South Korea, Germany

Article 6.4

- Countries can also trade units through a centralized mechanism overseen by A6.4 Supervisory Body
- Successor of Clean Development Mechanism
- Units issued under Article 6.4, called A6.4 ERs
    - AERs- Authorized A6.4 Emission Reduction Units
    - MCUs- Mitigation Contribution Units (unauthorized)
- Supervisory Body adopted many standards and procedures for operationalising A6.4
- Methodologies and its tools are under development

The Strategic Calculus

- ITMOs priced internationally vs. lower domestic CCTS
- Each ITMO exported shrinks India's NDC headroom
- Need: National Carbon Budget Framework — exportable surplus vs. domestic retention

Article 6 is not charity — it is carbon trade diplomacy. Opportunity to negotiate as a strategic supplier, not a default exporter of cheap credits.

Digital Infrastructure for Market Integrity: The Five Pillars

Without Digital Integrity, There Is No Market Integrity

Digital MRV

IoT-enabled CEMS, satellite verification, AI anomaly detection — the backbone of credit credibility

National Registry

Unique serialization, ownership tracking, audit trail of issuances and transactions

Interoperability

National registry must be ready for interoperating with UNFCCC A6.2 international registry, A6.4 mechanism registry and other national registries through API

Trading Platform

Leveraging IEX/PXIL/HPX infrastructure; spot → futures → full derivatives evolution

Data Transparency

Public dashboards, ISSB/BRSR alignment, greenwashing prevention through radical openness

The Road Ahead — Four Strategic Imperatives

From Market Launch to Market Maturity

Price Discovery & Liquidity

- Launch spot trading on IEX/PXIL/HPX with CCC instruments
- Introduce price floor/ceiling mechanism to prevent price collapse
- Incentivize market makers through fee waivers and spread guarantees
- Government auctioning of a portion of allowances for regular supply-side liquidity

Sectoral Expansion

- Expand beyond initial 282 industrial units to maximize coverage and market depth
- Priority sectors for Phase 2: Power generation, transport, buildings, agriculture, waste
- Include nature-based solutions to maximize impact and attract global buyer interest
- Every sector added = deeper liquidity + stronger carbon price signal

Article 6 Scale-Up

- Move beyond Japan JCM pilot → bilateral agreements with Switzerland, South Korea, EU member states
- Build institutional capacity for corresponding adjustments and NDC headroom management
- Develop a National Carbon Budget Framework — determine exportable surplus vs. domestic retention
- Position India as strategic carbon credit supplier, not default exporter of cheap credits

CBAM-ready Indian Industry

- Ensure CCTS sectoral coverage expands to all CBAM sectors — steel, aluminium, cement, fertilizers, hydrogen — and prepare for CBAM Phase 2 expansion to organic chemicals, plastics, and Scope 2 emissions by 2030
- Leverage FTA-based recognition as competitive advantage over exporters from non-FTA nations without domestic carbon pricing

India's carbon market is not just a climate instrument — it is simultaneously an industrial competitiveness strategy, a new financial asset class, and a tool of carbon trade diplomacy. The question is no longer 'should India have a carbon market?' — it is 'how fast can India make it deep, liquid, credible, and globally linked?

Thank you

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